SECTION ONE: Multiple-choice [50 marks]

This section has **25** questions. Answer **all** questions on the separate Multiple-choice Answer Sheet provided. For each question put a cross on the correct answer. If you make a mistake, strike out that answer and put a cross on another answer. Marks will not be deducted from your total for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time for this section is 30 minutes.

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1. The figure below shows the atomic symbol of element X:

$$^{15}_{7}\text{X}$$

Which of the following is the correct electron configuration for element X?

(a) 2, 5
(b) 2, 6
(c) 2, 7
(d) 2, 8, 5

2. A new element, olympium, was determined to have an atomic mass of 245.5. This is not a whole number because:

(a) The nucleus contains an extra half a neutron
(b) The element consists of several species containing different numbers of protons
(c) The element consists of several species containing different numbers of neutrons
(d) The element consists of several species containing different numbers of electrons

3. The table below shows information about particles A and B:

Particle	Proton number	Electron arrangement
A	11	2, 8 = 10 electrons
B	19	2, 8, 8 = 18 electrons

Based on the information provided, A and B are:

(a) Positive ions
(b) Negative ions
(c) Noble gases
(d) Isotopes of the same element

4. Which one of the following species does **NOT** have an electron configuration corresponding to an inert gas?

(a) Be^{2+}
(b) Rb^{+}
(c) At^{-}
(d) H^{+} — H^+ has no electrons

Questions 5 & 6 refer to the following information

The following represents a section of the periodic table. The elements shown have fictitious symbols, and are non-metals; none are inert gases.

A			J
		G	
E			

greater electroneg. and I.E.

bigger radius

5. Which element would have the smallest atomic radius?
(a) A
(b) E
(c) J
(d) G

6. Which element would have the lowest electronegativity?
(a) A
(b) E
(c) J
(d) G

7. All of the following properties except one increase as one considers the elements of a typical chemical family in order, starting at the top of a column of the periodic table and going down. Which property does **NOT** increase going down the periodic table?

(a) Mass number
(b) Atomic weight
(c) Number of valence electrons — same within a group.
(d) Number of filled orbitals

8. A compound is formed when element Y (atomic number = 12) is heated with element Z (atomic number = 9). The compound formed would be:

Mg

F

(a) Ionic, formula YZ_2
(b) Covalent, formula YZ
(c) Ionic, formula YZ
(d) Covalent, formula YZ_2

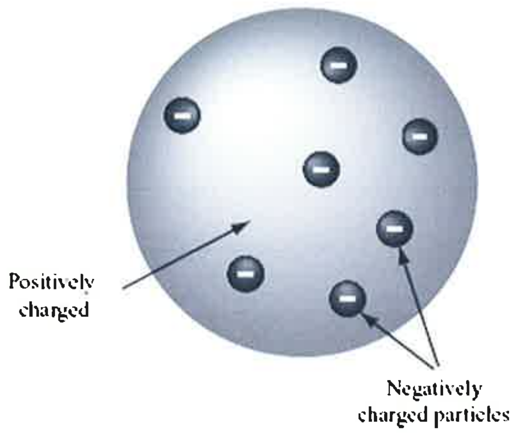
MgF_2, ionic.

9. Consider an ion of uranium, $^{235}_{92}U^{3+}$. In this ion there are:

(a) 92 neutrons
(b) 89 protons
(c) 89 electrons
(d) 95 electrons

→ Means 92 protons.
But it has a charge of +3
so must have lost 3 electrons.
92 − 3 = 89. electrons.

<u>Questions 10 & 11 refer to the following model of an atom</u>



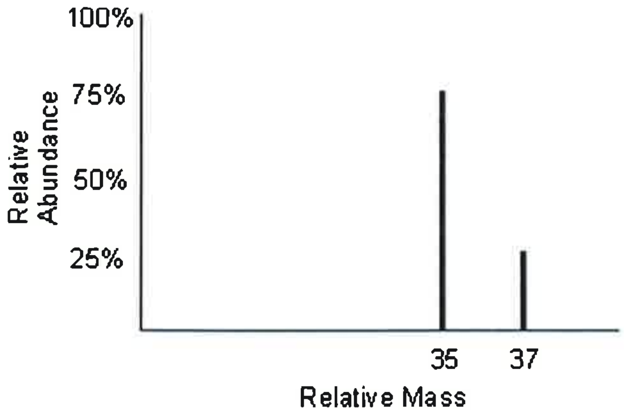
10. Which scientist was responsible for this model of the atom?

(a) Niels Bohr
(b) Ernest Rutherford
(c) J. J. Thomson
(d) John Dalton

11. What scientific discovery lead to the development of this model?

(a) Elements combine to form compounds in fixed ratios
(b) Alpha particles pass through gold foil undeflected
(c) Cathode rays are made of negatively charged particles
(d) Emission spectra of hydrogen consists of discrete lines

Questions 12 & 13 refer to the following diagram



12. What is the name for this type of diagram?

(a) Line graph
(b) Calibration curve
(c) Chromatogram
(d) Mass spectrum

13. Why are there two separate peaks on the diagram?

(a) The atom emits two different wavelengths of light
(b) There are two compounds in the mixture
(c) The atom was split in half during its analysis
(d) There are two isotopes of the element – (the element is chlorine. Peaks are Cl-35 and Cl-37)

14. The formula of a chloride salt of M is $MC\ell_2$. What is the formula of a sulfate salt of M?

M^{2+} Cl^-

M^{2+} SO_4^{2-}

(a) MSO_4
(b) M_2SO_4
(c) $M(SO_4)_2$
(d) $M_2(SO_4)_2$

15. What is the formula of the hydrogenphosphate ion?

(a) HPO_4^{2-}
(b) HPO_4^-
(c) $H_2PO_4^-$
(d) $H_2PO_4^{2-}$

16. Which of the following pairs of elements forms a compound by sharing electrons?

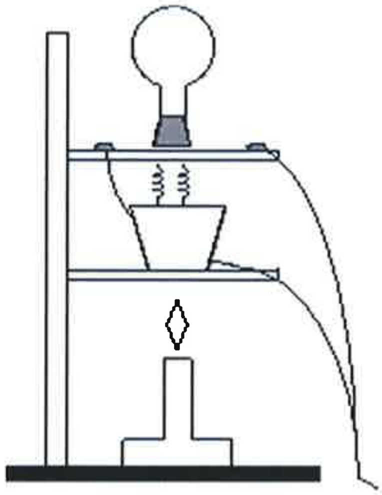
Covalent

(a) Sulfur and lead
(b) Magnesium and oxygen
(c) Nitrogen and bromine — (two non-metals)
(d) Potassium and iodine

Questions 17 & 18 refer to the following information

The conductivity of solid and molten substances can be tested using the apparatus shown below. A crucible containing the solid is heated with a Bunsen burner until it melts. Two electrodes extend from the light bulb into the crucible. If the substance conducts electricity then current flows through the light bulb and the globe turns on.

Two substance (**X** and **Y**) were tested using this apparatus and the results of the experiment are provided in the table.



Substance	Melting point	Electrical conductivity	
		Solid state	Liquid state
X	620 °C	Poor	Good – Ionic
Y	48 °C	Poor	Poor – Covalent

low melting point = covalent molecular

17. What type of compound matches the properties of **Substance X**

(a) Covalent molecular
(b) Covalent network
(c) Ionic
(d) Metallic

18. What type of compound matches the properties of **Substance Y**

(a) Covalent molecular
(b) Covalent network
(c) Ionic
(d) Metallic

19. How many moles of oxygen are required to react completely with 2.5 moles of methane to form water?

$\underset{2.5\,mol.}{CH_4(g)} + \underset{?}{2O_2(g)} \rightarrow CO_2(g) + 2H_2O(g)$

(a) 1.25 mol
(b) 0.625 mol
(c) 2.50 mol
(d) 5.00 mol

20. Which quantity is the same for one mole of water and one mole of nitrogen gas?

(a) The number of atoms
(b) The mass
(c) The volume at room temperature
(d) The number of molecules

21. Which one of the following statements about chemical reactions is **NOT** true in all cases?

(a) The mass of the reactants consumed equals the mass of the produced formed
(b) No atoms are created or destroyed in the reaction
(c) The number of moles of products equals the number of moles of reactants
(d) The reactant produces new substances through the rearrangement of atoms

Can have e.g. $\underbrace{2H_2 + O_2}_{3\text{ mol reactants}} \rightarrow \underbrace{2H_2O}_{2\text{ mol products.}}$

22. Which of the following equations is correctly balanced?

(a) $Cu(OH)_2 \rightarrow CuO + 2\ H_2O$
(b) $A\ell_2(SO_4)_3 + 4\ NaOH \rightarrow 2\ A\ell(OH)_3 + 2\ Na_2SO_4$
(c) $SnO_2 + 2\ H_2 \rightarrow Sn + 2\ H_2O$
(d) $Na_2SO_3 + 4\ HC\ell \rightarrow 2\ NaC\ell + SO_2 + 2\ H_2O$

23. Which formula would calculate the number of water molecules in 50 g of water (molar mass 18.016 g mol^{-1})?

(a) $N(H_2O) = \frac{50}{18.016} \times 6.022 \times 10^{23}$

(b) $N(H_2O) = \frac{18.016}{50} \times 6.022 \times 10^{23}$

(c) $N(H_2O) = \frac{50}{18.016 \times 6.022 \times 10^{23}}$

(d) $N(H_2O) = \frac{18.016}{50 \times 6.022 \times 10^{23}}$

$N = n \times 6.022 \times 10^{23}$

and $n = \frac{m}{M}$

so $N = \frac{m}{M} \times 6.022 \times 10^{23}$

24. Propane combusts according to the following reaction:

$$\overset{0.688}{} \quad ?$$

$$C_3H_8 + 5\ O_2 \rightarrow 3\ CO_2 + 4\ H_2O$$

Which of the following would calculate the number of carbon dioxide molecules produced from 0.688 moles of oxygen gas?

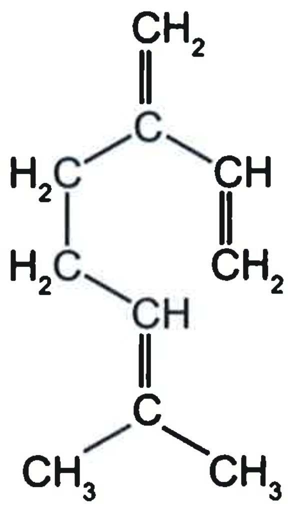
(a) $n(CO_2) = \frac{5}{3} \times 0.688$

(b) $n(CO_2) = \frac{3}{5} \times 0.688$

(c) $n(CO_2) = \frac{5}{3 \times 0.688}$

(d) $n(CO_2) = \frac{3}{5 \times 0.688}$

25. Myrcene is an organic compound found in the essential oils of sweet basil, hops, mangoes and cannabis. It's full chemical structure is shown below:



Molecular: $C_{10}H_{16}$

Empirical : C_5H_8

(simplest ratio)

What is the empirical formula of myrcene?

(a) $C_{10}H_{16}$
(b) $C_{10}H_7$
(c) C_5H_8
(d) CH_2

END OF SECTION ONE

SECTION TWO: Short Answer [70 marks]

This section has **15** questions. Answer **all** questions. Write your answers in the space provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time for this section is 70 minutes.

Question 26 **(6 marks)**

(a) Complete the table by writing the formula of each chemical species. (3 marks)

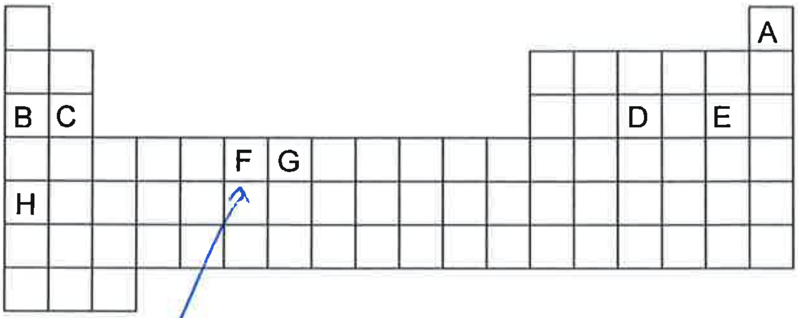
Name	Chemical formula
Sulfur hexachloride	SCl_6
Calcium hydrogensulfate	$Ca(HSO_4)_2$
Hydrogen peroxide	H_2O_2

(b) Complete the table by writing the name of each chemical species. (3 marks)

Chemical formula	Name
PbO	Lead (II) oxide
H_3PO_4	Phosphoric acid
NBr_3	Nitrogen tribromide

Question 27 **(10 marks)**

The periodic table is arranged carefully to display many of the trends present in the 92 naturally occurring elements found on Earth. Answer the following questions referring only to the elements labelled.



(a) Which element could have the following subatomic particle arrangement? (1 mark)

protons = 24 neutrons = 31 electrons = 24

F

(b) Which 2 elements would be most similar in their chemical properties? (1 mark)

B and H (same group)

(c) Write the electron configuration of element D. (1 mark)

2, 8, 5 OR $1s^2 2s^2 2p^6 3s^2 3p^5$

(d) Which element is most likely to form an ion with a charge of +2? (1 mark)

C

(e) Which element is diatomic? (1 mark)

E

(f) Which 2 elements would have the same number of valence electrons? (1 mark)

B and H

(g) Write the formula and name of the most likely substance formed when C and E combine. (2 marks)

CE_2 or $MgCl_2$. Magnesium chloride.

(h) Write the formula and name of the most likely substance formed when D and E combine (2 marks)

DE_3 or PCl_3. Phosphorus trichloride.

Question 28 **(2 marks)**

Complete the following table to compare the relative masses and charges of protons, neutrons and electrons.

Subparticle	Relative mass	Relative charge
Proton	1	+1
Neutron	1	0 / None / Neutral
Electron	~ 1/2000	−1

Question 29 **(6 marks)**

Write balanced ionic equations for each of the following reactions. If no reaction occurs, state "no reaction" for the equation. For each scenario describe what is observed, including the colours of any solutions and precipitates.

(a) Barium chloride solution is mixed with a sodium sulfate solution. Solid barium sulfate and aqueous sodium chloride solution are produced. (3 marks)

Molecular equation: $BaCl_{2(aq)} + Na_2SO_{4(aq)} \rightarrow BaSO_{4(s)} + 2NaCl_{(aq)}$

Ionic equation: $Ba^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)}$

(b) A drop of hydrochloric acid is added to solid sodium carbonate. Water and aqueous sodium chloride are produced, along with carbon dioxide gas. (3 marks)

Molecular equation: $Na_2CO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(g)} + CO_{2(g)}$

Ionic equation: $Na_2CO_{3(s)} + 2H^+_{(aq)} \rightarrow 2Na^+_{(aq)} + H_2O_{(l)} + CO_{2(g)}$

Question 30 **(4 marks)**

One of the isotopes of aluminium is $^{27}_{13}\mathrm{A}\ell$.

Complete the following sentences to describe information about this isotope.

(a) The name of this isotope is aluminium-27 (1 mark)

(b) The mass number is 27 and the atomic number is 13 (1 mark)

(c) The atom has 13 protons and 14 neutrons in the nucleus (1 mark)

(d) The electron configuration of the atom is 2, 8, 3 (1 mark)

(or $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$)

Question 31 **(4 marks)**

For each of the species listed below, draw the structural formula, representing all valence shell electron pairs either as : or as –

Water has been done as an example.

Species	Structure (showing all valence shell electrons)
Water (H_2O)	·· H — O — H ··
Carbon dioxide (CO_2)	 1 mark for double bonds 1 mark for lone pairs.
Potassium bromide (KBr)	 1 mark for ionic structure 1 mark for $[\ ]^+[\ ]^-$

Question 32 **(3 marks)**

Name the type(s) of bonding (ionic, covalent ~~molecular~~, ~~covalent network~~ or metallic) present in each of the following substances:

Substance	Type(s) of bonding
Graphite	Covalent (network)
Mercury	Metallic
Chlorine	Covalent (molecular)
Magnesium sulfate	Ionic and covalent. (Ionic bonds join Mg^{2+} to SO_4^{2-} but covalent bonds join S atoms to O atoms)
Rubidium chloride	Ionic
Hydrogen bromide	Covalent (molecular)

* The "network" and "molecular" refers to their structure, but wasn't needed for this question

Question 33 **(4 marks)**

Classify each of the following substances by listing them in the correct column of the table below.

- Chlorine
- Tap water
- Solder *
- Vinegar – mixture of acetic acid and water.
- Blood
- $NaC\ell(aq)$
- $Al_2O_3(s)$
- $S_8(s)$

**Note: Solder is an alloy used in electronics. It is made by melting lead and tin together.*

Pure substance		Mixture	
Element	Compound	Homogeneous	Heterogeneous
Chlorine	$Al_2O_3(s)$	Tap water	Blood
$S_8(s)$		Solder	
		Vinegar	
		NaCl(aq)	

0.5 marks each

Question 34 (6 marks)

(a) List the mobile charge carriers that are responsible for conducting electricity in each of the following substances or solutions. (4 marks)

Substance	Mobile charge carriers
K(s)	delocalised electrons.
$MgBr_2(\ell)$	Mg^{2+} ions and Br^- ions
Graphite	delocalised electrons
$Cu(\ell)$	delocalised electrons and Cu^{2+} ions

(b) Explain why magnesium bromide does **not** conduct electricity when it is in the solid state. (2 marks)

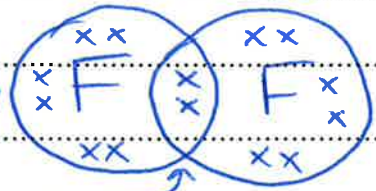
1 • Ions in fixed network lattice

1 • Ions Cannot move so cannot act as "mobile charge carriers"

Question 35 (3 marks)

Using fluorine gas (F_2) as an example, explain the difference between a "shared electron pair" and a "lone electron pair".

• Fluorine gas consists of two fluorine atoms joined by a covalent bond. 1



• Shared pair: Electrons shared between both F atoms so they have a full outer shell. 1

• Lone pairs: Valence electrons belonging to one F atom only. 1

Question 36 **(4 marks)**

(a) Calculate the number of moles of water molecules in 6.00 g of water. (2 marks)

$M(H_2O) = 18.016$ g/mol. 1

$n(H_2O) = m/M$

$= 6.00 / 18.016$

$= 0.333$ mol. 1

(b) Calculate the number of moles of methane in 1.6 x 10^{24} molecules of methane. Give your answer to three significant figures. (2 marks)

$n(CH_4) = \frac{1.6 \times 10^{24}}{6.022 \times 10^{23}}$ 1 for answer

$= 2.67$ mol. 1 for three sig. figures.

Question 37 **(2 marks)**

Water vapour is formed when hydrogen burns in oxygen.

(a) Write a balanced molecular equation for this reaction. (1 mark)

$2H_2 + O_2 \rightarrow 2H_2O$.

(b) Determine the number of moles of oxygen required for the complete combustion of 4.00 moles of hydrogen. (1 mark)

~~$n(H_2) =$~~ $n(O_2) = 1/2 \times n(H_2)$

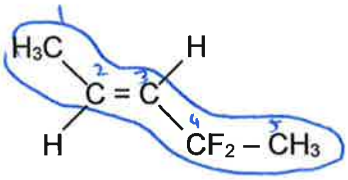
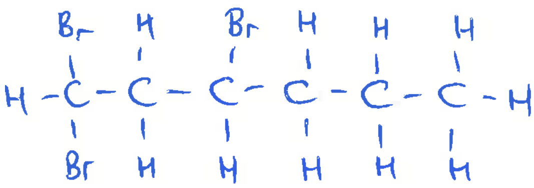
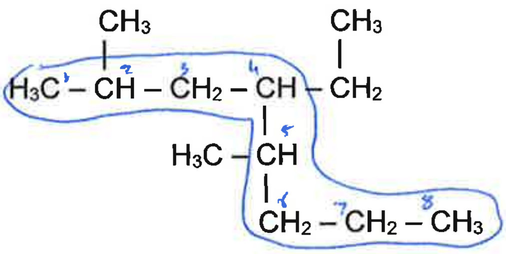
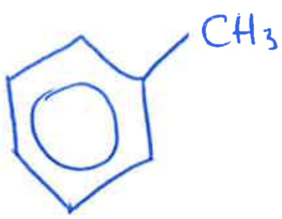
$= 1/2 \times 4.00$

$= 2.00$ mol 1

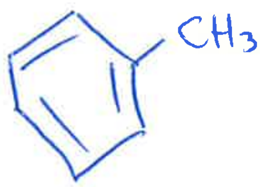
Question 38

(4 marks)

Complete the following table by either naming the molecule using the IUPAC system or drawing a structural diagram of the molecule.

	IUPAC Name	Structural Diagram
A	4,4-difluoropent-2-ene	
B	1,1,3-tribromohexane	
C	4-ethyl-2,5-dimethyloctane	
D	methylbenzene	

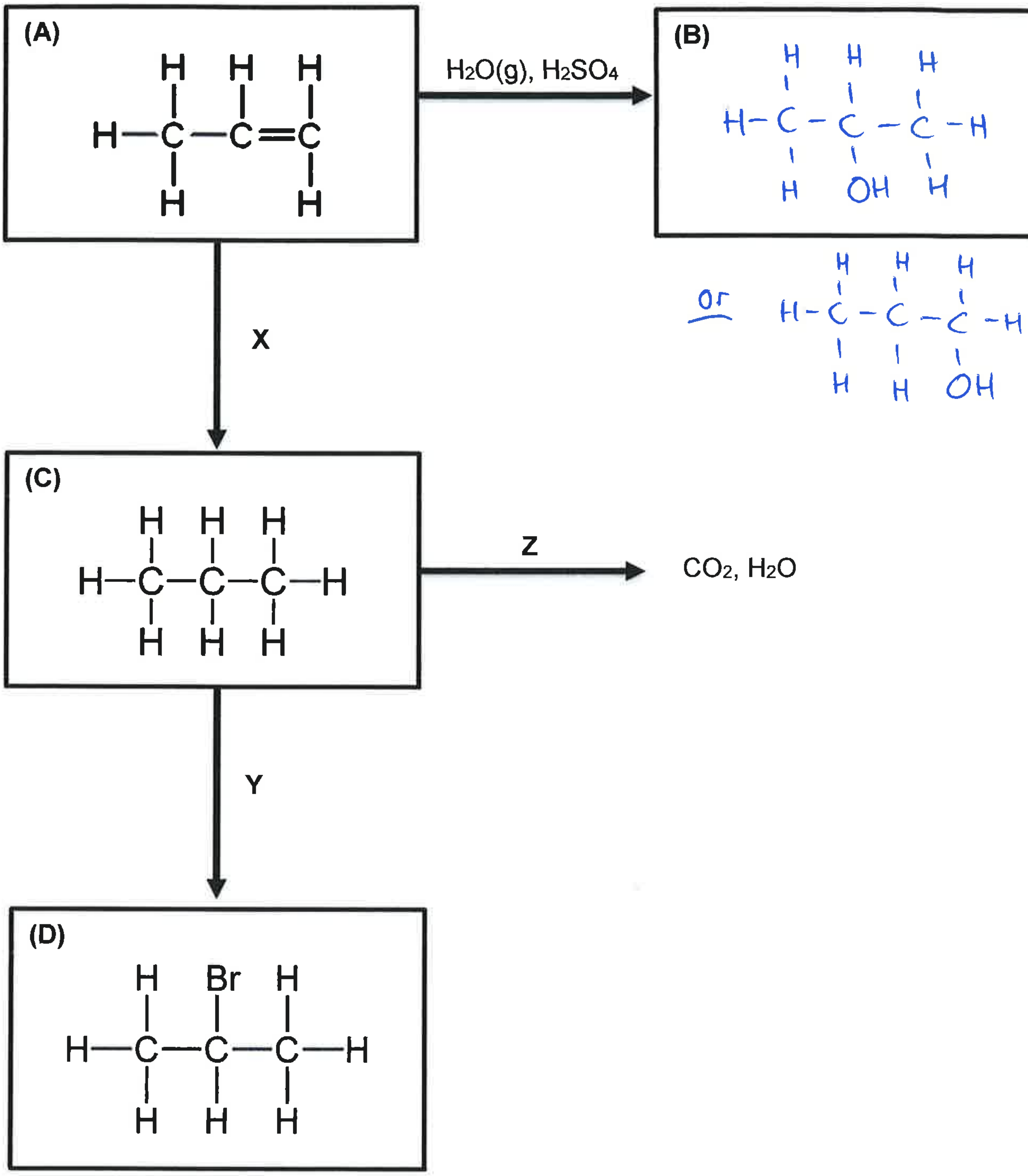
or



or with H's shown.

Question 39 **(9 marks)**

Refer to the following diagram when answering questions on the next page.



(a) Name compound **A**. (1 mark)

propene

(b) Draw the structure of **B**, the molecule produced after reacting **A** with $H_2O(g)$ in the presence of sulfuric acid catalyst. (1 mark)

(c) What chemical reacted with **A** to form **C**? (1 mark)

H_2

(d) Name the type of reaction occurring a **X** (1 mark)

Addition

(e) What chemical reacted with **C** to form **D**? (1 mark)

Br_2

(f) Name the type of reaction occurring a **Y**. (1 mark)

Substitution

(g) Name compound ~~C~~ D (1 mark)

2-bromopropane.

(h) Write a balanced full molecular equation representing reaction Z. (1 mark)

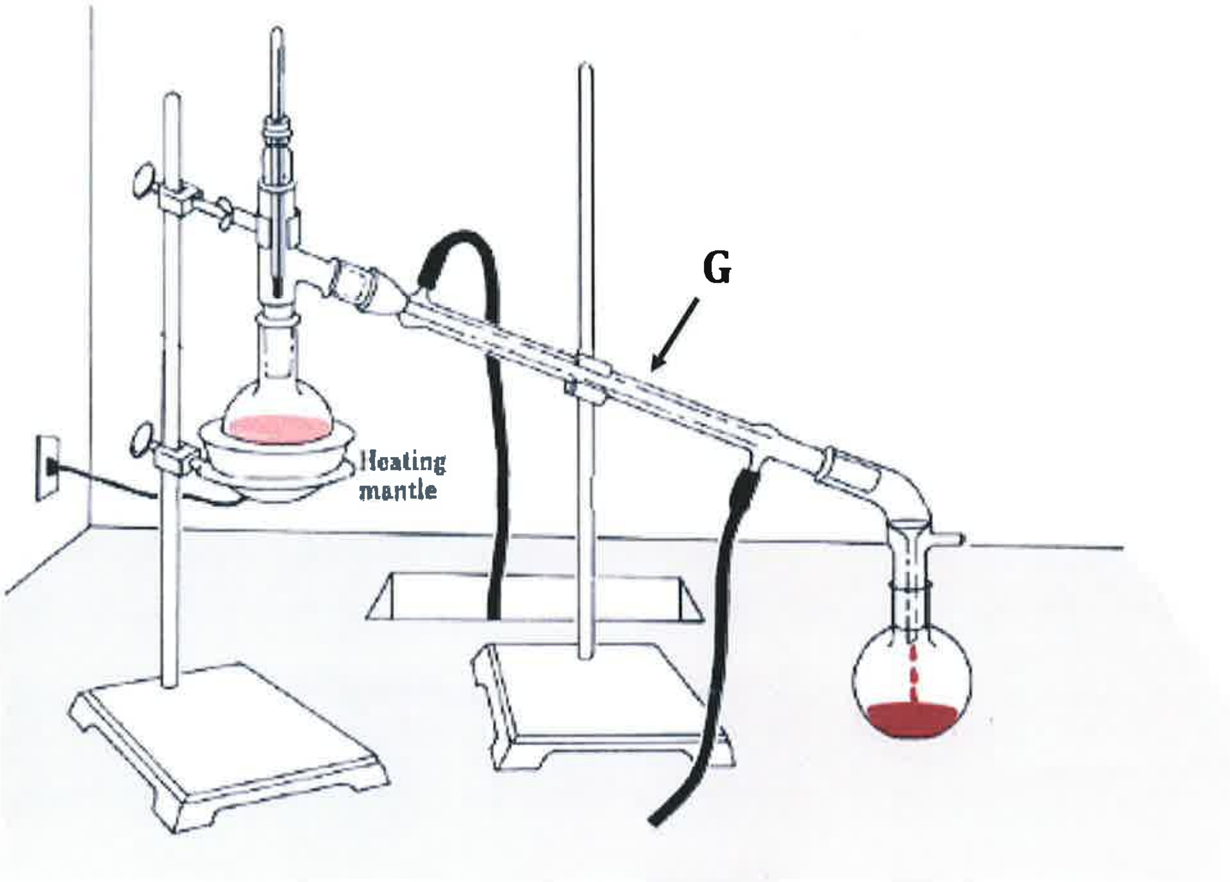
$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

(i) Under what conditions would carbon monoxide have been produced instead of carbon dioxide in reaction Z? (1 mark)

Limited amount of oxygen.

Question 40 **(3 marks)**

The pictures apparatus is a method for separating substances with different boiling points.



(a) Name this separation technique. (1 mark)

Distillation.

(or "simple distillation")

(b) Give the name of the glass tube labelled "G". (1 mark)

Condenser

(c) Briefly describe the function of "G". (1 mark)

• Keeps tube cold so that vapourised samples will condense into liquids for collection.

END OF SECTION TWO

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Section 3 begins on the next page.

SECTION THREE: Extended Answer [80 marks]

This section contains **six (6)** questions. You must answer **all** questions. Write your answers in the space provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time for this section is 80 minutes.

Question 41 **(19 marks)**

(a) The page to the right shows a rotated version of the periodic table. Elements in Groups 1, 2, 13, 15, 16 and 17 form charged ions with either a negative or a positive charge.

Fill in the boxes on the page to the right to show the charge formed by ions in each group (e.g. "**2+**"). (2 marks)

- 1 mark for the positive ions
- 1 mark for the negative ions.

(b) In terms of electron configuration, what property do all elements in a group of the periodic table share? (1 mark)

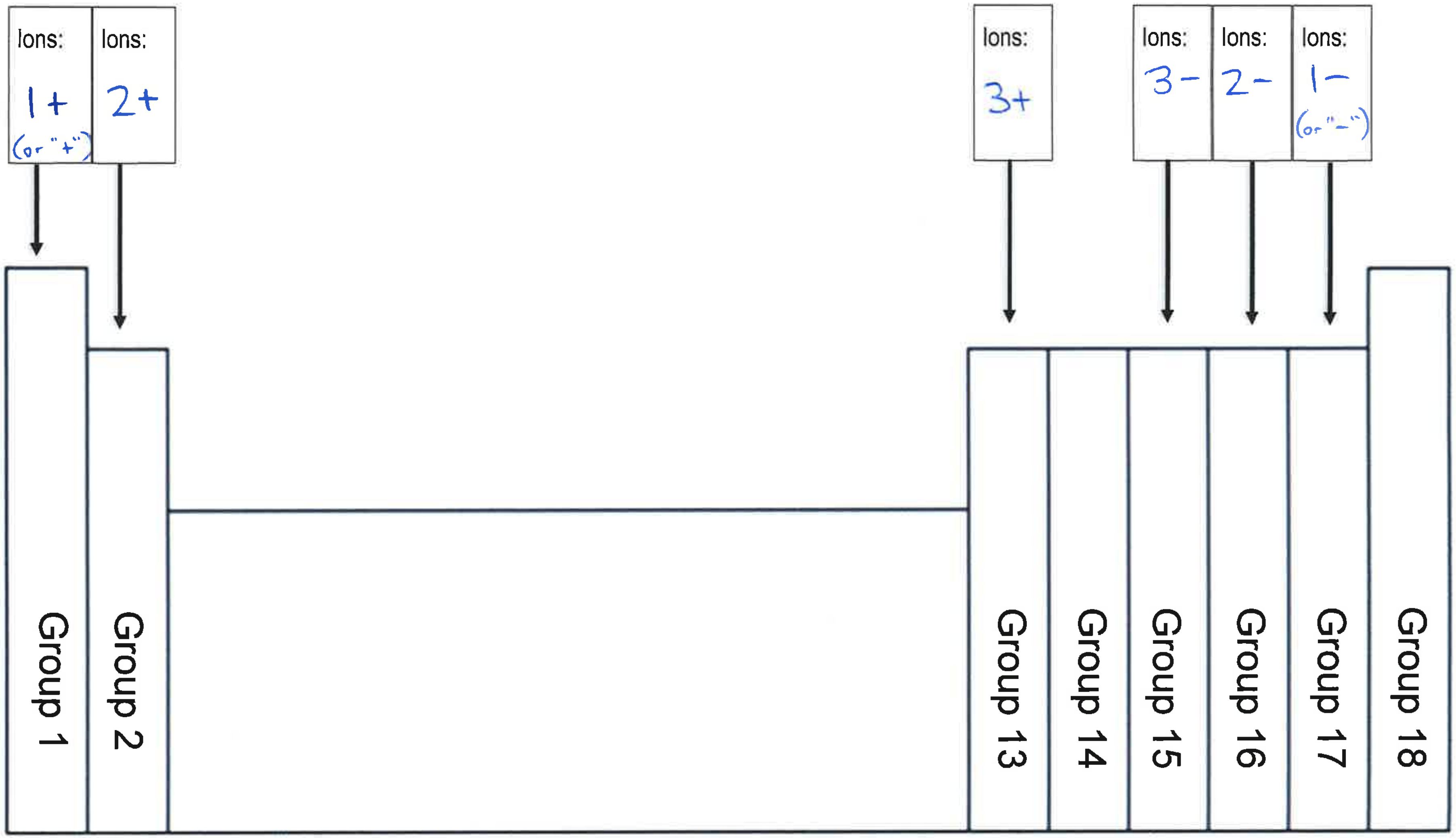
same number of valence electrons.

(c) Explain how and why all elements in Group 17 form the particular charge of ion that they form. (3 marks)

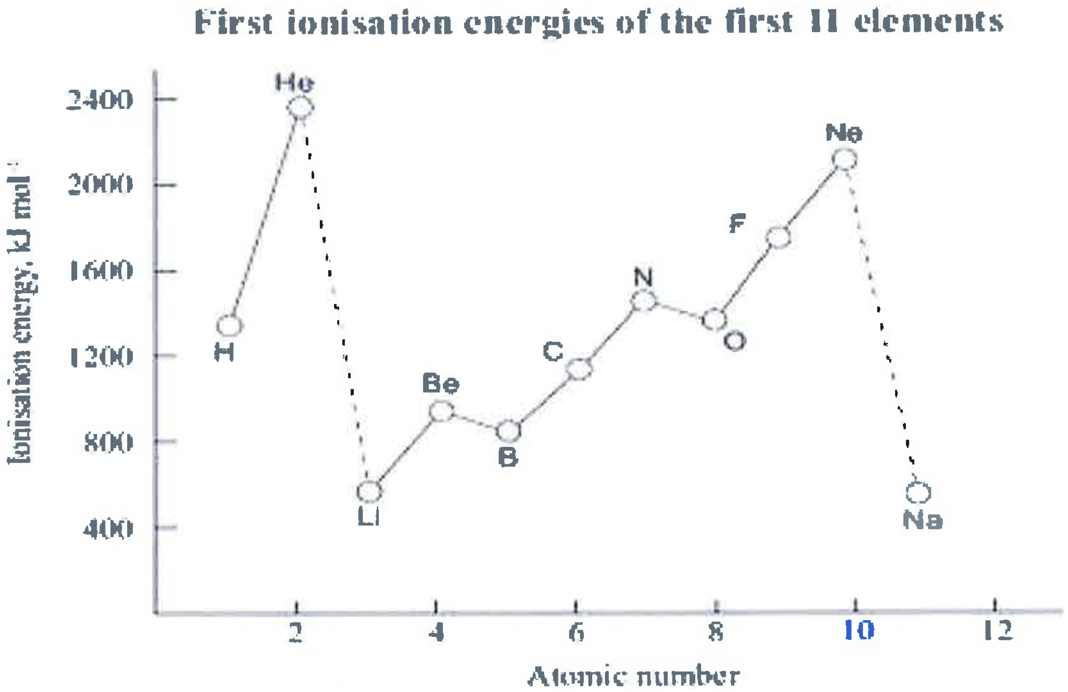
- All have 7 valence electrons. 1
- All gain one electron... 1

... so they can gain a full outer shell of valence electrons 1

Diagram for Question 41a



(d) Describe and account for the trends observed on the following graph: (6 marks)



Trend going across a period:

• Going across, ionisation energy increases. (1)

• There are more protons in the nucleus... (1)

... so valence electrons experience a stronger force of attraction to nucleus } (1)

Trend going down a group:

• Going down, ionisation energy decreases. (1)

• Electrons are added to outer shells... (1)

... and the inner electrons shield some of the positive charge from the nucleus. } (1)

(e) Fluorine forms an ionic compound with sodium atoms, but forms a covalent compound with oxygen atoms.

i. Describe what happens when fluorine forms an ionic bond with sodium (2 marks)

There is a transfer of electrons 1

Na loses electron to form Na^+
F gains electron to form F^- 1

ii. Describe what happens when fluorine forms a covalent bond with oxygen (2 marks)

F and O share electrons... 1

... so each has a full outer shell 1

iii. Explain in terms of electronegativity and ionisation energy why fluorine forms different types of compounds with oxygen and sodium respectively. (3 marks)

- Fluorine has high electronegativity 1
- Sodium has low ionisation energy (loses electron easily) so fluorine takes the electron. 1
- Oxygen has high ionisation energy and high electronegativity. Neither atom can take electron from other so they share. 1

Question 42 **(12 marks)**

Analysis of a sample of an organic gas was found to contain 23.8% carbon (C), 5.90% hydrogen (H) and 70.3% chlorine (Cℓ) by mass.

(a) Calculate the empirical formula of the organic gas. (6 marks)

	C	H	Cl	
Mass (in 100g)	23.8 g	5.90 g	70.3 g	← 1
moles	1.98 mol	5.85 mol	1.98 mol.	← 3
Mole ratio.	$\frac{1.98}{1.98}$ = 1	$\frac{5.85}{1.98}$ = 3	$\frac{1.98}{1.98}$ = 1	← 1

Empirical formula is CH_3Cl. ← 1

(b) Further analysis of the sample revealed that 0.115 moles of the organic gas had a mass of 5.82 g. Calculate the molar mass of the compound, including correct units. (2 marks)

$n = m/M$

$\therefore M = m/n.$

$M = 5.82 / 0.115$

$= 50.6 \text{ g mol}^{-1}$ 1 for number

1 for units.

(c) Using your answers to parts (a) and (b);

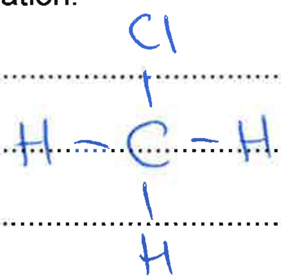
(i) Show that the molecular formula of the gas is the same as its empirical formula. (2 marks)

$M(CH_3Cl) = 12.01 + 3 \times 1.008 + 35.45$

$= 50.484 \text{ g mol}^{-1}$ 1

This is approximately equal to molar mass from part b, therefore molecular formula is CH_3Cl. 1

(ii) Draw and name a possible structure for this organic molecule which matches all of the provided information. (2 marks)

Structure:  1

Name: chloromethane. 1

Question 43 13 ~~(12~~ **marks)**

Baking powder is used as a raising agent and is often listed as an ingredient in cake recipes. 'Single acting' baking powder consists of one leavening agent, such as sodium bicarbonate (sodium hydrogencarbonate, $NaHCO_3$). This is mixed with an acidic substance such as calcium dihydrogenphosphate monohydrate ($Ca(H_2PO_4)_2H_2O$, as well as other ingredients such as corn starch or sodium aluminium sulfate.

The equation below shows a simplified version of the leavening process, involving the reaction between sodium bicarbonate and calcium dihydrogenphosphate monohydrate. The reaction releases carbon dioxide gas, which in turn causes the cake to rise.

$$14\ NaHCO_3 + 5\ Ca(H_2PO_4)_2H_2O \rightarrow Ca_5(PO_4)_3OH + 7\ Na_2HPO_4 + 18\ H_2O + 14\ CO_2$$

(a) According to the law of conservation of mass, matter is not created or destroyed in chemical reactions. Despite this, the mass of a cooked cake decreases during the cooking process. Why does the mass of the cake appear to decrease? (2 marks)

• Forms $CO_2\ (g)$ 1

• The $CO_2\ (g)$ escapes, so mass of cake decreases. 1

* (Also, $H_2O(l)$ would evaporate)

(b) One of the compounds formed during the leavening process is Na_2HPO_4. Give the name for this compound. (1 mark)

sodium hydrogen phosphate. 1

(c) Calculate the molar mass of the two reactants. 2 (~~1~~ mark) 5

Substance	Molar mass
$NaHCO_3$	84.008 $g\ mol^{-1}$
$Ca(H_2PO_4)_2H_2O$	252.068 $g\ mol^{-1}$

(d) If there is an excess of sodium hydrogencarbonate ($NaHCO_3$) in the reaction mixture then the baked cake may still have significant amounts of sodium hydrogencarbonate left. This can produce an undesirable salty flavour.

What mass of sodium hydrogencarbonate should be combined with 2.10 g of calcium dihydrogenphosphate monohydrate so that both reactants are completely used with no reactants left over? (3 marks)

mass known $m(Ca(H_2PO_4)_2H_2O) = 2.10$ g.

↓

moles known $n(Ca(H_2PO_4)_2H_2O) = m/M$

$= 2.10/252.068$

$= 0.008331$ mol. 1

↓

moles unknown $n(NaHCO_3) = \frac{14}{5} \times n(Ca(H_2PO_4)_2H_2O)$

$= 0.02333$ mol. 1

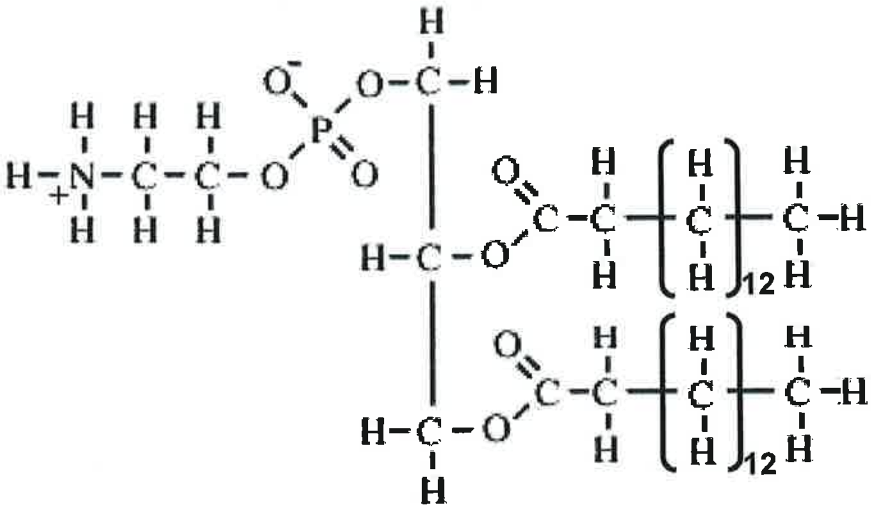
↓

mass unknown $m(NaHCO_3) = n \times M$

$= 0.02333 \times 84.008$

$= 1.96$ g. 1

Another compound that plays a key role in cake baking is **lecithin**. Licithin is an emulsifier and a surfactant, which means it helps to dissolve substances such as fats and oils in water. Licithin is used in cake mixtures so that water stirs in easily and to improve the consistency of the mixture.



Structure of lecithin

(e) Give the molecular formula of lecithin in the form of CvHwOxNyPz. (2 marks)

$C_{35}H_{70}O_8NP$.

-1 per mistake.

(f) Calculate the percentage by mass of carbon in lecithin. Give your answer to three significant figures. (3 marks)

$M(C_{35}H_{70}O_8NP) = 663.89 \text{ g mol}^{-1}$ 1

$\% C = \frac{35 \times 12.01}{663.89} \times 100$

$= 63.3\%$ 1

+ 1 for 3 sig figures.

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Questions continue on the next page.

Question 44 **(16 marks)**

Diborane, B_2H_6, can be used as a rocket fuel due to the fact that it ignites easily with air. It is a colourless gas with a sickly sweet odour and can cause skin and eye irritation and shortness of breath if inhaled. There are many methods that can be used to prepare diborane and one of the most common is shown below. This reaction has a yield of about 30%.

$$\underline{4}\ BF_3\ (g) \ + \ \underline{3}\ NaBH_4\ (s) \ \rightarrow \ \underline{2}\ B_2H_6\ (g) \ + \ \underline{3}\ NaBF_4\ (s)$$ 1

(a) Balance the chemical equation above. (1 mark)

The combustion equation for diborane, as would occur in the fuel tank of a rocket, is shown below.

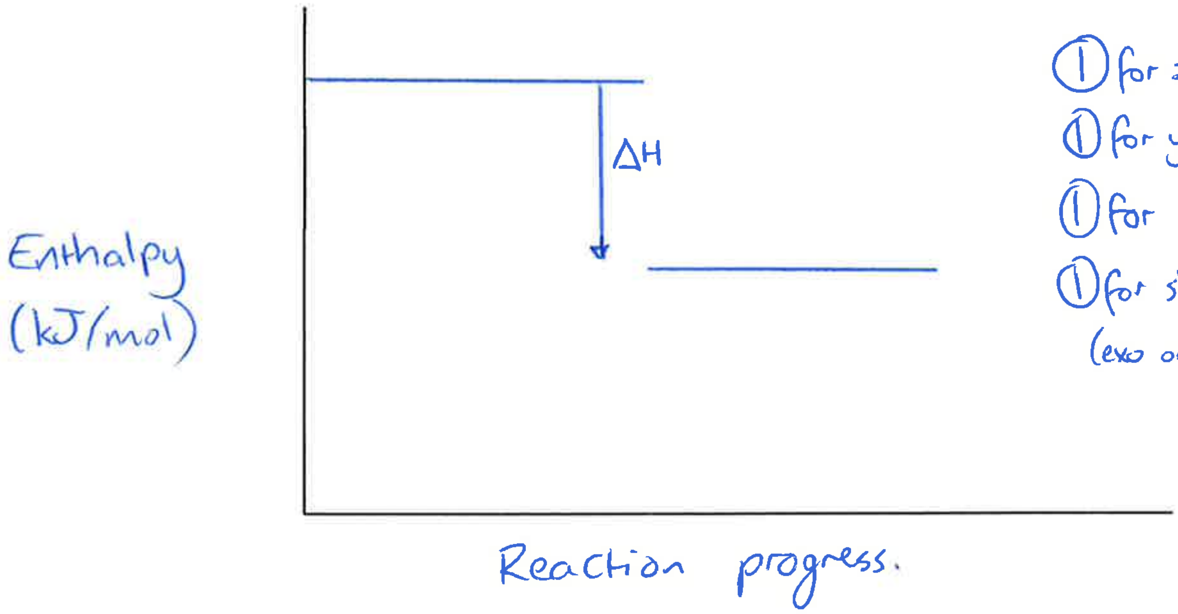
$$B_2H_6\ (g) \ + \ 3\ O_2\ (g) \ \rightarrow \ B_2O_3\ (s) \ + \ 3\ H_2O\ (l) \ + \ 2168\ kJ$$

(b) Is the combustion of diborane endothermic or exothermic? (1 mark)

Exothermic 1

(Allow follow-through marks for rest of question if "endo-")

(b) Sketch a labelled potential energy diagram for the combustion of diborane. Label the axes and ΔH. (4 marks)



1 for x-axis label
1 for y-axis label
1 for ΔH arrow
1 for shape of graph (exo or endothermic)

(c) State the value of the following; (2 marks)

ΔH of the forwards reaction: $\Delta H = -2168\ kJ/mol$ 1

ΔH of the reverse reaction: $\Delta H = +2168\ kJ/mol$ 1

(d) Explain what information the heat of reaction (ΔH) for the combustion of diborane provides in terms of the bond-breaking and bond-making processes that occur.

(3 marks)

· Bond-making releases energy (1)

· Bond-breaking absorbs energy (1)

· Given that the overall reaction was exothermic, more energy was released during bond-making than consumed during bond breaking. (1)

for then making comparison of energy released/absorbed.

(d) Calculate the amount of energy absorbed or released from the combustion of 1 kg of diborane (B_2H_6). Give your answer to three significant figures. (3 marks)

~~1000~~ 1 kg = 1000 g of B_2H_6...

$n(B_2H_6) = m/M$

$= 1000/27.688$

$= 36.1$ mol. (1)

Reaction releases 2168 kJ per mole.

∴ 36.1 moles releases $2168 \times 36.1 = 78264.8$ kJ (1)

$= 7.82 \times 10^4$ kJ (1)

(to 3 sig. figures)

(e) Most fuels used in everyday society are fossil fuels, however researchers are looking into the potential of using alternative fuel sources such as biogas, biodiesel and ethanol. Select **one** alternative fuel source and list **one advantage** and **one disadvantage** compared to fossil fuels. (2 marks)

Alternative fuel source	*Circle one…* Biogas Biodiesel Ethanol
Advantage of the alternative fuel source	....
Disadvantage of the alternative fuel source	....

Biodiesel

- Advantages: - can be used in existing cars
  - safer than diesel (higher flashpoint)
  - better for environment / carbon-neutral
- Disadvantages: - Releases less energy
  - Cost of production.

Biogas:

- Advantages: - made from waste material
  - can be compressed
- Disadvantages: - cost of production
  - can't be used in existing vehicles.

Ethanol:

- Advantages: - Less CO_2 output than petrol
- Disadvantages: - Less energy output than petrol
  - Cost of production.

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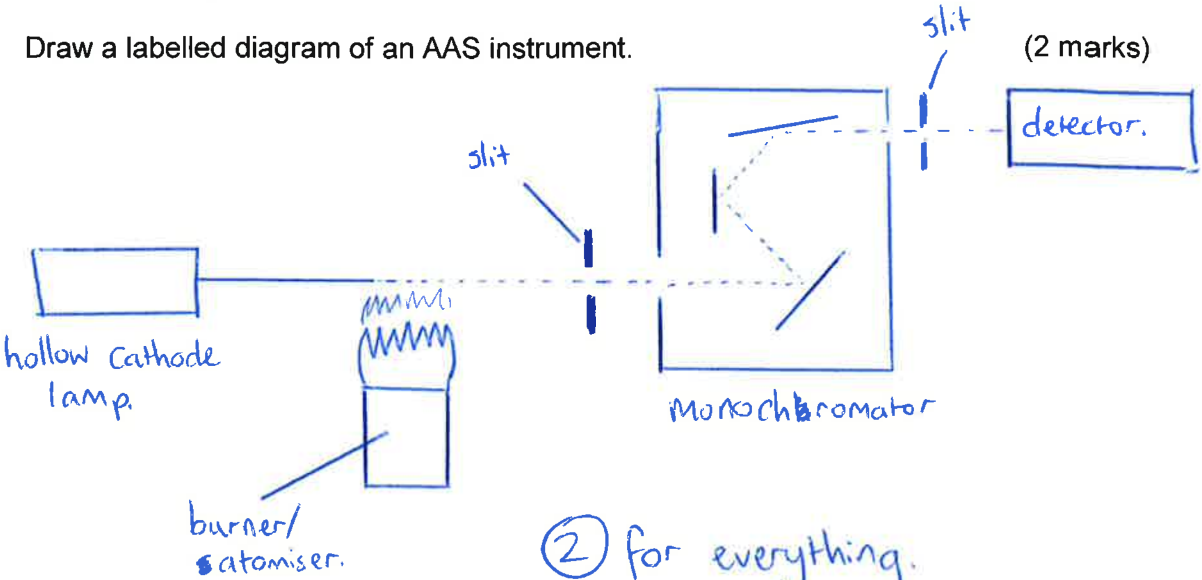
Questions continue on the next page.

Question 45 **(10 marks)**

Atomic absorption spectroscopy (AAS) is an analytical technique which can measure the concentration of heavy metals.

(a) Draw a labelled diagram of an AAS instrument. (2 marks)

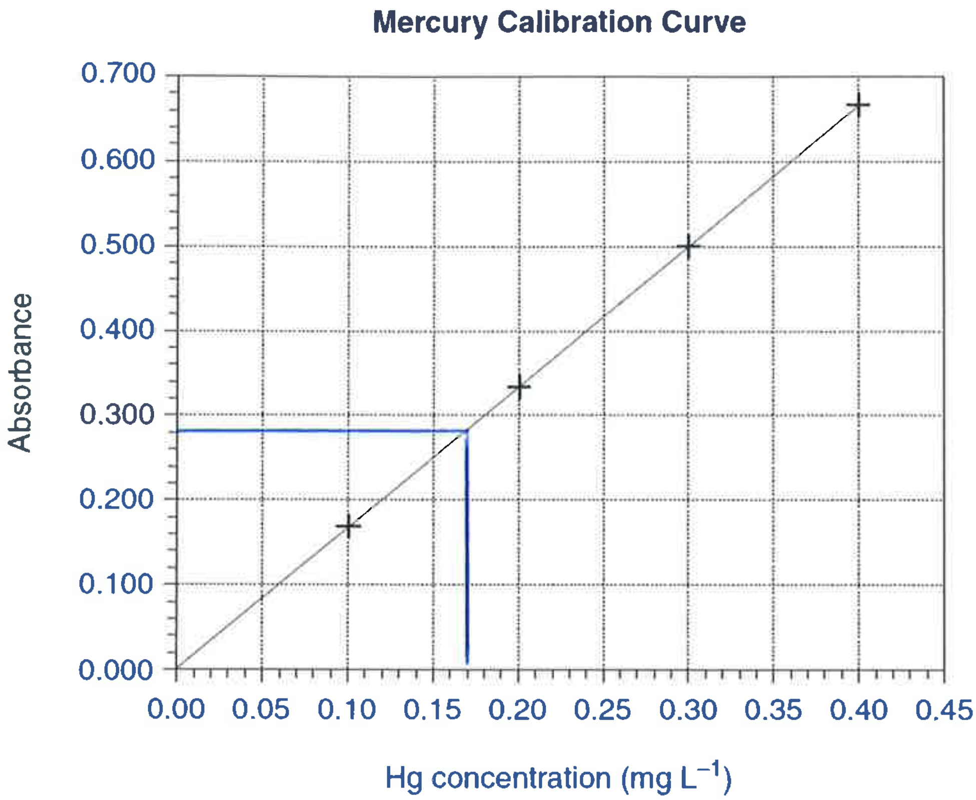


② for everything.
- ½ mark for each thing missing
(treat both slits as one "thing")

(b) **Emission** and **absorption** are two processes that occurring during atomic absorption spectroscopy. Complete the table below to show the differences between these two processes. (4 marks)

	Emission	Absorption
What occurs during the process?	Electrons relax to ground state and emit light.	Electrons absorb light to enter excited state.
Where in AAS does it occur?	Hollow cathode lamp.	In /Near the flame.

The mercury concentration of a certain fish species was determined by atomic absorption spectroscopy. 18.6 g of fish was used to prepare 25 mL of sample solution. The absorbance of the sample solution was 0.280.



(c) What was the concentration of mercury in the fish sample? (1 mark)

0.17 mg/L 1

(d) What was the mass (in mg) of mercury present in the **25 mL** sample solution? (1 mark)

25 mL = 0.025 L. 1

1 L has 0.17 mg, ∴ 0.025 L has... 0.025 × 0.17 = 0.00425 mg.

(e) A consumer wants to avoid eating fish with a mercury concentration greater than 0.5 mg/kg of fish. Show using a calculation whether the fish is safe. (2 marks)

18.6 g fish = 0.0186 kg fish.

$\frac{0.00425 \text{ mg Hg}}{0.0186 \text{ kg fish}}$ = 0.228 mg/kg fish. 1

therefore safe to eat. 1

Question 46 **(10 marks)**

Oxygen is one of the most abundant elements on Earth. It makes up 21% of the air we breathe and can be found in an almost unending number of different compounds such as water, sand, rust, DNA and proteins. Since oxygen is such a ubiquitous substance, most elements have the ability to react with it. The table below shows three different oxides and their boiling points.

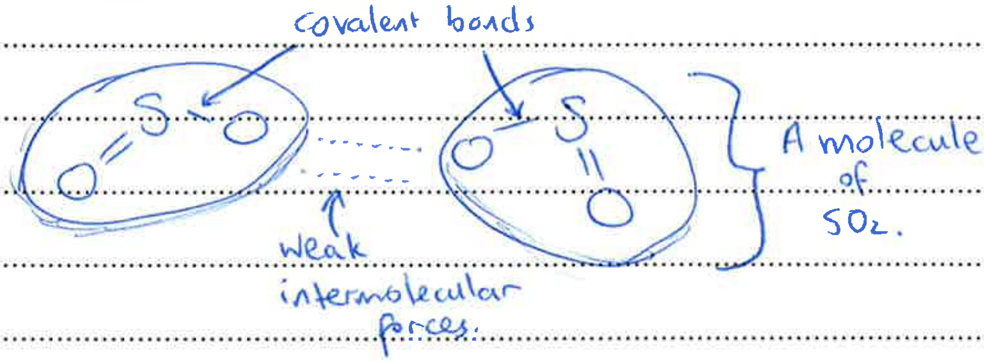
	SO_2	MnO_2	SiO_2
Boiling point (°C)	-10	535	2230

In terms of structure and bonding present, explain the differences in boiling point of the three oxides. Use diagrams in your answer where appropriate. Your answer should be one to two pages in length.

1 mark for good use of diagrams throughout.

SO_2:

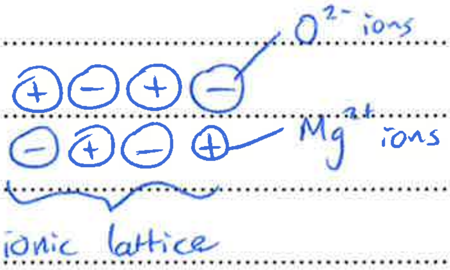
- SO_2 is a covalent molecular substance. 1 Inside each molecule there are strong covalent bonds joining atoms but only weak intermolecular forces joining each molecule. 1



- The low boiling point of SO_2 can be explained by the weakness of the intermolecular forces joining molecules. Only a small amount of heat energy is required to break these weak forces. 1

MnO_2:

- MnO_2 is an ionic substance. 1
- It exists as a 3D network lattice of alternating positive and negative ions. There is a strong force of electrostatic attraction between the oppositely charged ions. 1



- MnO_2 has a high ~~melting~~ boiling point because it requires a large amount of energy to disrupt the strong ionic bonds. 1

SiO_2:

- SiO_2 is a covalent network substance. 1
- All atoms a joined by covalent bonds, which is a strong form of electrostatic attraction between the positive nuclei and shared electrons. 1
- SiO_2 has an extremely high ~~melting~~ boiling point because covalent bonds are very strong and require a very large amount of energy to break. 1

END OF EXAMINATION